WGCNA Analysis of lncRNA Expression Data: Pipeline Structure, Results, and Signed vs. Unsigned Network Comparison

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Contents

1 Pipeline Structure

The lncRNA script (`lncRNA_WGCNA.R`) follows the exact same stage order as the mRNA pipeline, so this section only covers the parameters and logic that differ from the mRNA run.

1.1 Global Parameters

- `MIN_COUNT = 5`, `MIN_FRAC = 0.50`. Less strict compared to the mRNA script's `MIN_COUNT = 10`. lncRNAs are far more less expressed than protein-coding mRNAs, so applying the mRNA threshold would have discarded a disproportionate share of lncRNA signal.
- `N_GENES_WGCNA = NA`, `MAX_BLOCK_SIZE = 12000`, `N_HUBS = 20`.

1.2 Loading and Size-Factor Reuse

The file (`lncRNA_filter1.xlsx`) is the same 963-sample cohort with 4115 lncRNA count columns plus `Subtype`. As with the mRNA script, DESeq2 is run with an intercept-only design ($\sim$ `1`) purely as a normalisation/VST tool. Critically, **size factors are not re-estimated here**: they are ***imported directly from the mRNA run's saved `mrna_size_factors.rds`, matched to the lncRNA cohort by sample name (with an explicit check that all 963 samples have a match before proceeding)***.

Justification: the lncRNA count matrix has a substantially higher zero fraction than the mRNA matrix (17.2% vs. 5.1%), and library depth is a property of the sequencing run rather than of transcript biotype, so the size factors estimated from the denser mRNA counts are the more reliable per-sample depth correction for both data types.

2 Results

2.1 Cohort and Preprocessing Summary

The cohort is identical to the mRNA run: 963 samples, PAM50 subtype counts Basal 170, HER2 75, LumA 528, LumB 190. Of the 4115 raw lncRNA genes, only **2300 (55.9%) passed the low-expression filter**, **compared to 83.1% for mRNA at the stricter threshold**. Per-sample lncRNA library totals (median 250,574; range 62,624–625,909) are roughly two orders of magnitude smaller than the mRNA library totals. **All 963 samples and all 2300 genes entered network construction**.

2.2 MAD Distribution

Figure 1 shows the same right-skewed shape seen in the mRNA MAD histogram: most of the 2300 filtered genes cluster at low-to-moderate MAD (roughly 1.0–1.5), with a long thin tail of more variable genes extending out past MAD = 4–6. As in the mRNA run, `N_GENES_WGCNA` was left at `NA`, so no MAD cutoff was applied and all 2300 filtered genes were carried into WGCNA.

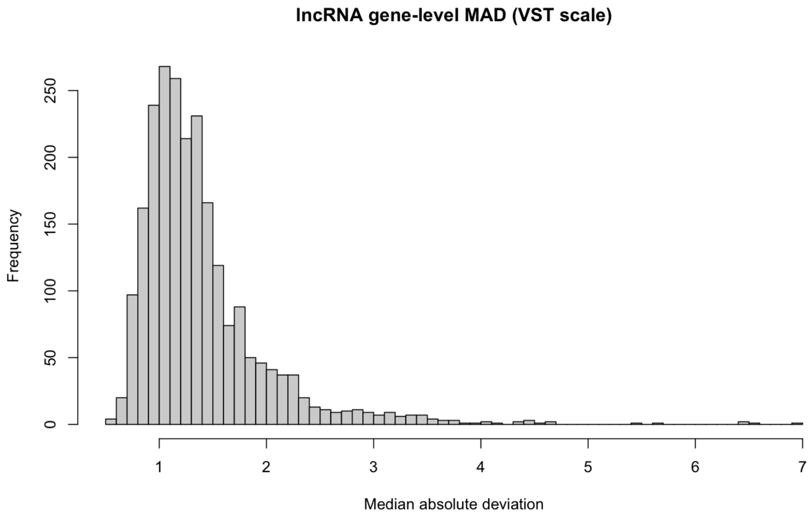

Figure 1: Distribution of gene-level MAD (VST scale) across the 2300 expression-filtered lncRNA genes.

2.3 Sample Clustering

Figure 2 shows the same style of outlier-screening dendrogram used for mRNA. No sample or small cluster of samples sits at a dramatically greater height than the rest of the tree, so no cut height was applied and all 963 samples were retained.

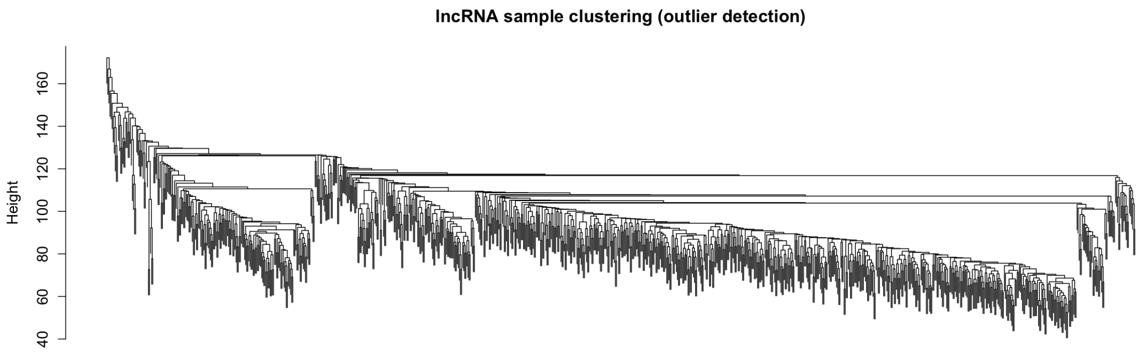

Figure 2: Average-linkage hierarchical clustering of all 963 samples on lncRNA expression (outlier screening). No cut was applied.

2.4 Soft-Threshold Selection

Figure 3 and Table 1 summarise power selection. The signed network's fit curve rises smoothly and monotonically, crossing $R^2 = 0.90$ at power 14 ($R^2 = 0.931$). The **unsigned** fit curve, by contrast, is unstable at higher powers: it rises smoothly to $R^2 = 0.945$ at power 7, then collapses to $R^2 \approx 0.36$ at power 10, partially recovers to $R^2 = 0.960$ at power 12, and collapses again ($R^2 < 0.40$) from power 14 onward. The automatic power-selection rule (lowest power at which R^2 first reaches 0.90) is unaffected by this instability, since it correctly picks the first, stable crossing at power 5 ($R^2 = 0.924$).

Table 1: Selected soft-thresholding power and fit statistics for each lncRNA network type.

Network type	Chosen power	Scale-free R^2 at chosen power	Mean connectivity
Unsigned	5	0.924	1.81
Signed	14	0.931	1.51

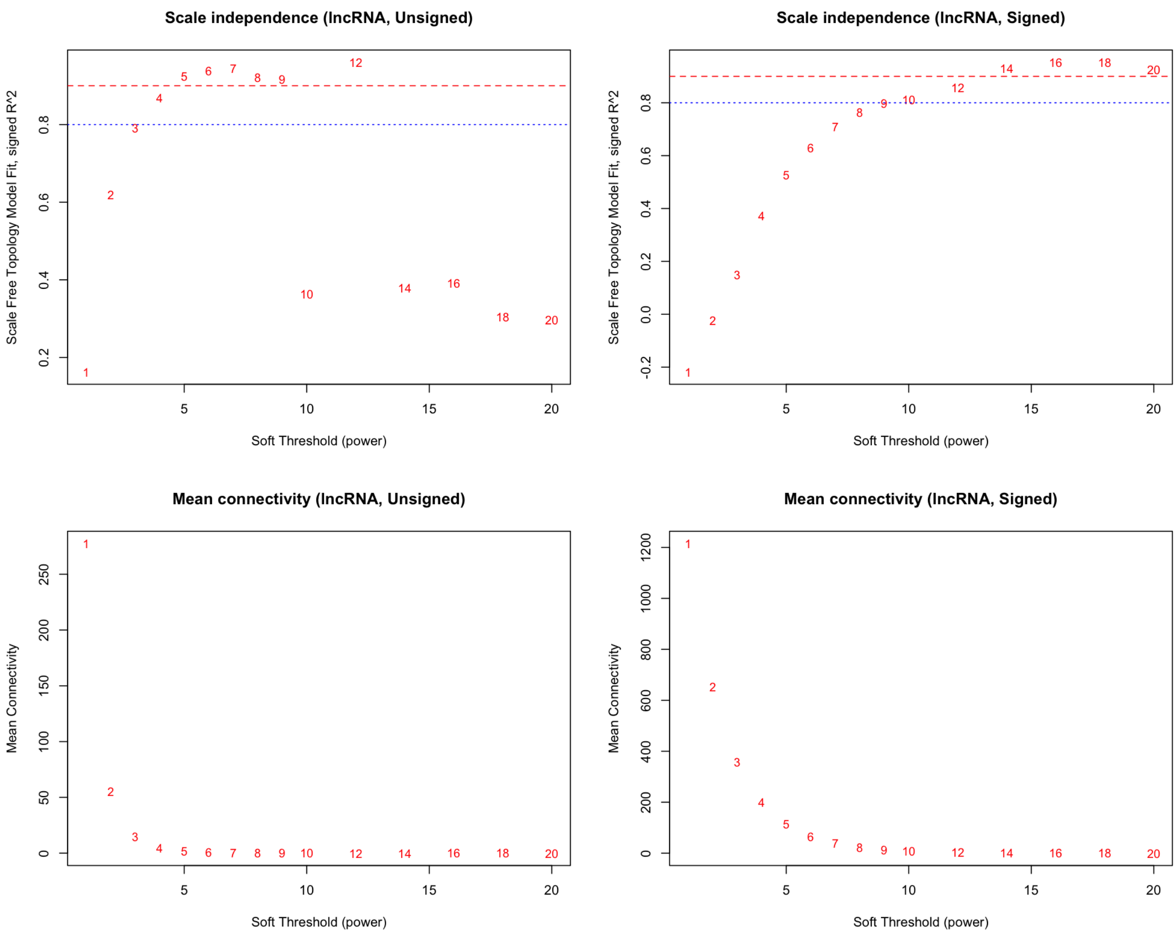

Figure 3: Scale-free topology fit and mean connectivity vs. soft-thresholding power for the unsigned (left) and signed (right) lncRNA networks. Note the unsigned fit's instability above power 9.

2.5 Network Construction and Module Detection

Table 2: lncRNA module sizes under both parameter sets. "First run (used)" is the assignment used throughout this report, chosen for its better-resolved module structure; "Second run" was also computed but not carried forward, since it merged several first-pass modules together.

	Signed		Unsigned	
Module	First run (used)	Second run	First run (used)	Second run
black	41	—	53	—
blue	201	207	282	291
brown	83	83	115	107
green	54	—	84	74
magenta	—	—	44	—
pink	34	—	51	—
red	50	—	74	52
turquoise	217	225	283	298
yellow	82	54	106	84
grey (unassigned)	1538	1731	1208	1394
Total non-grey	762 (33.1%)	569 (24.7%)	1092 (47.5%)	906 (39.4%)
Total genes	2300	2300	2300	2300

The **unsigned** network organises a larger number of lncRNA genes into non-grey modules than the signed network (47.5% vs. 33.1% first-pass; 39.4% vs. 24.7% tuned), the opposite of what was observed for mRNA. **However, in my opinion, I still prefer the signed network for reasons mentioned further**.

Figure 5 shows the gene dendrograms for both network types under the first-pass parameters chosen for this analysis.

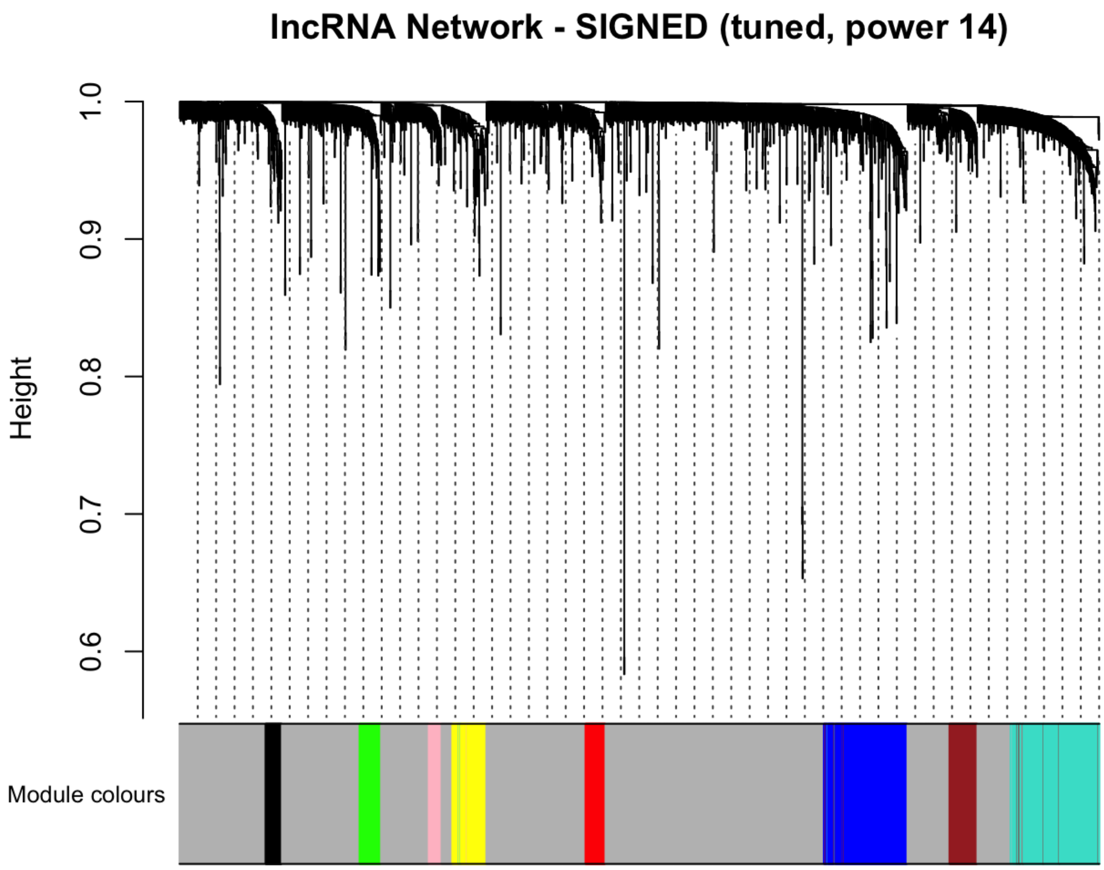

Figure 4: Gene clustering dendrogram with module colours, signed lncRNA network (power 14, first-pass parameters). Module colour bands are largely solid, contiguous blocks corresponding to single well-separated branches.

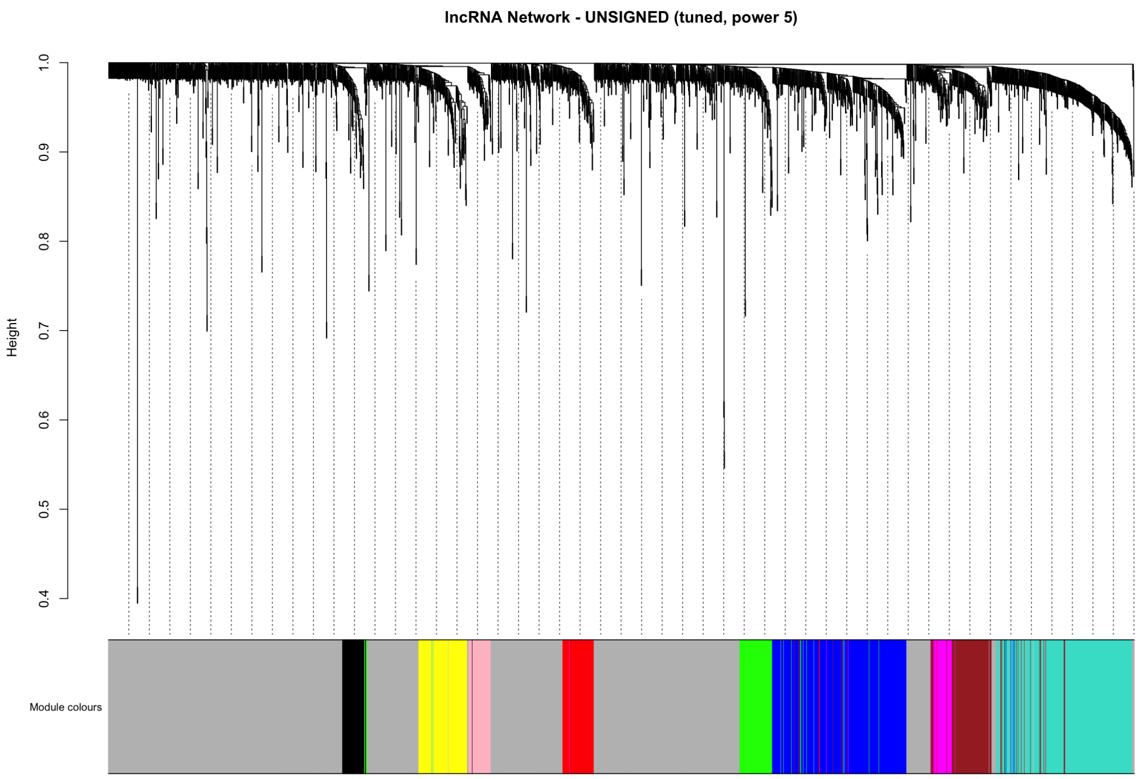

Figure 5: Gene clustering dendrogram with module colours, unsigned lncRNA network (power 5, first-pass parameters). Several module colours (e.g. yellow, blue, turquoise) appear as multiple separated stripes rather than one contiguous block.

2.6 Module Eigengene Correlation Structure

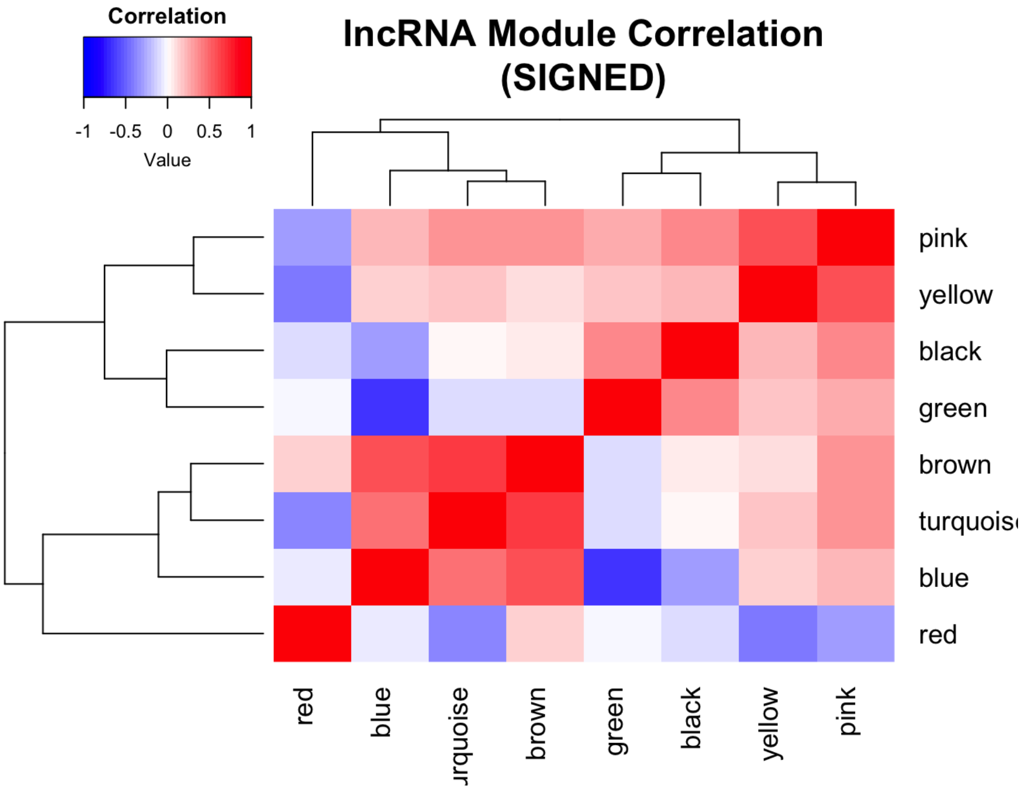

Figure 6: Module eigengene correlation heatmap, signed lncRNA network.

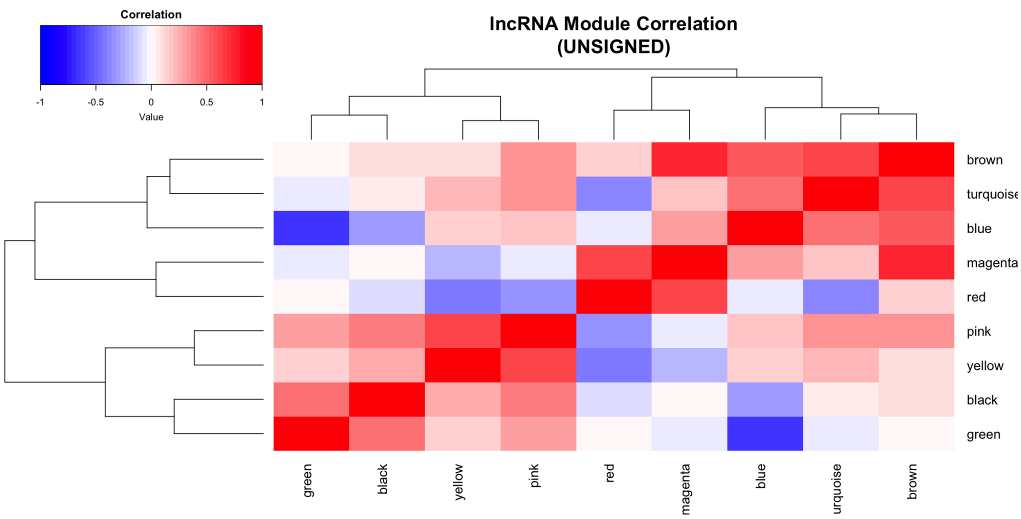

Figure 7: Module eigengene correlation heatmap, unsigned lncRNA network.

2.7 Module–Trait Relationships

Table 3 and Figure 8 report the signed module–subtype correlations; Figure 9 reports the unsigned equivalent. As with mRNA, the dominant pattern is a Basal/LumA opposition, led here by `blue` ($r = -0.84$ Basal, $r = 0.61$ LumA) and `green` ($r = 0.76$ Basal). ***Unlike the mRNA result, HER2 shows more modules with a detectable association here (e.g. brown $r = -0.21$, blue $r = -0.21$), and LumB shows a clearer distinguishing signal***

in a subset of modules (pink $r = -0.29$***, green*** $r = -0.41$***)*** that is largely independent of the Basal/LumA axis, consistent with the third-axis interpretation of the eigengene clustering above.

Table 3: Signed module–subtype Pearson correlations, lncRNA network.

Module	Basal	HER2	LumA	LumB
blue	−0.84	−0.21	0.61	0.17
green	0.76	−0.11	−0.19	−0.41
brown	−0.31	−0.21	0.38	−0.04 (n.s.)
turquoise	−0.26	−0.12	0.24	0.03 (n.s.)
yellow	−0.23	−0.01 (n.s.)	0.32	−0.17
pink	−0.21	−0.11	0.46	−0.29
black	0.19	0.10	−0.08	−0.14
red	0.15	0.03 (n.s.)	−0.15	0.02 (n.s.)

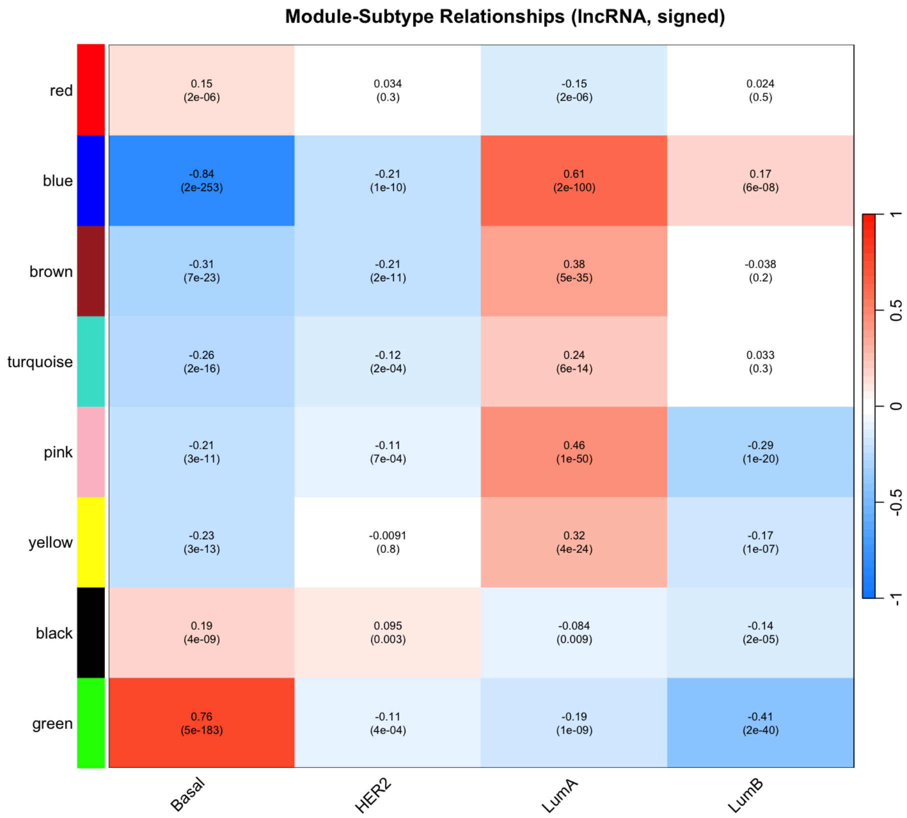

Figure 8: Module–subtype relationships, signed lncRNA network. Each cell reports Pearson r (top) and p-value (bottom).

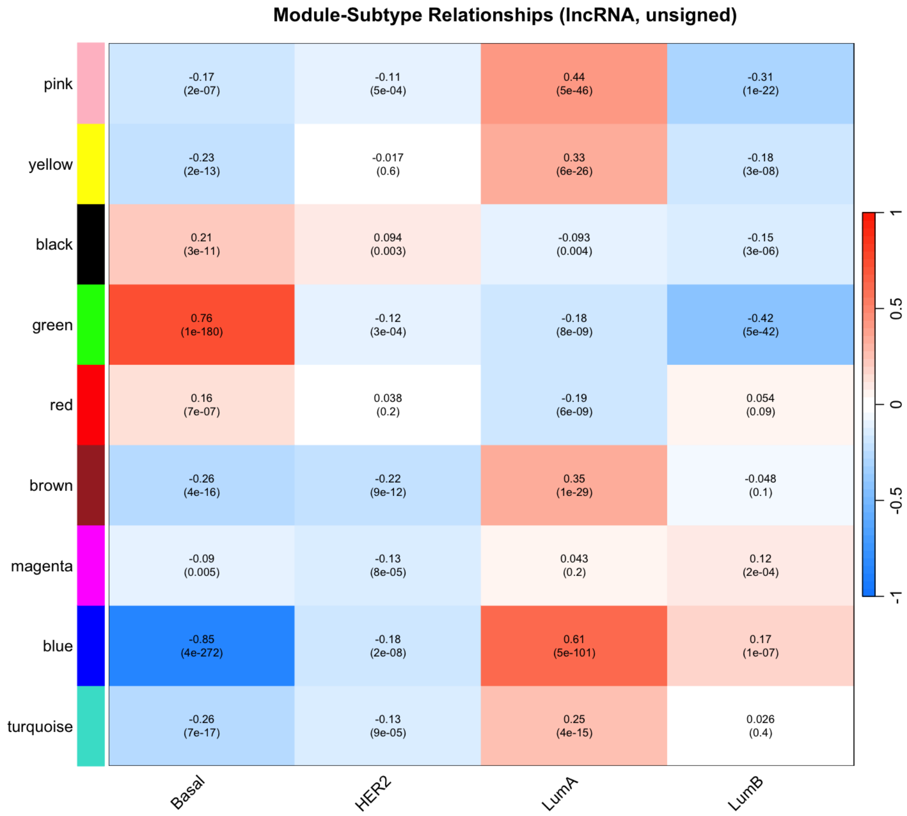

Figure 9: Module–subtype relationships, unsigned lncRNA network. Each cell reports Pearson r (top) and p-value (bottom).

2.8 Hub Genes

The top kME hub gene per signed module was: blue – *LINC00504* (kME 0.869); green – *LINC02188* (0.848); turquoise – *NPTN-IT1* (0.858); black – *PCED1B-AS1* (0.846); brown – *RAD51-AS1* (0.846); yellow – *TBX5-AS1* (0.836); pink – *LINC01197* (0.809); red – *AL121832.2* (0.856). No gene appeared in more than one module in either network. Notably, and unlike the mRNA result, six of these eight top hub genes are identical between the signed and unsigned networks at the *same* colour label (*LINC00504*/blue, *LINC02188*/green, *NPTN-IT1*/turquoise, *PCED1B-AS1*/black, *RAD51-AS1*/brown, *AL121832.2*/red); only `yellow` (signed: *TBX5-AS1*; unsigned: *MAGI2-AS3*) and `pink` (signed: *LINC01197*; unsigned: *LINC02884*) differ, and the unsigned network additionally resolves an extra `magenta` module (top hub *AC092119.2*) with no signed counterpart. This is a different pattern from the mRNA result, where signed and unsigned hub genes were scrambled across colour labels.

3 Signed vs. Unsigned for lncRNA

Reasons I think we should go with signed here also:

1. **Directionality is guaranteed by construction, not merely observed.** Signed adjacency only connects genes that are *positively* correlated with each other, so every gene inside a signed module is guaranteed to move in the same direction as every other member; the module eigengene's sign then directly means "this whole module goes up in X and down in Y," with no need to check individual member genes.

2. **The signed soft-thresholding fit is also more stable.** The unsigned fit curve degrades sharply at higher powers (Section 2.4, Figure 3), while the signed fit rises smoothly and monotonically.